Time For Informed Change.

Post Covid-19 Economics

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Abstract.

In these extraordinary times it might just be that heterodox economics gets a hearing; if only to justify government actions ruled impossible or, at least undesirable, by mainstream economics in normal times; it is back to the 2008 future all over again, big-time. So, if economics were to descend from its 'theological heights' (and preaching only that which suits elite vested interests), then what are we to say? This article utilises alternative theoretical lenses to underpin views of fiscal and monetary policy and the case for state banking. It also expresses an opinion as to which capital is worth saving, post-crisis. More generally, we consider if advanced nations should aim to be more self-sufficient in the future and if so, how might developing countries fit into a new order? We are not prophets or salespeople, so we merely seek to provide some economic theory that can help us understand these issues. The theories we apply are Modern Monetary Theory and the Temporal Single System Interpretation of Marx (which argues that his value theory is consistent and not redundant, in fact invaluable to understanding capitalism). Space will not permit us to drag up endless *academic* debates on the *acceptability* of the TSSI of Marx or MMT, and for once, in the face of crisis, we hope we may be spared from abiding by the rules of the club.[1]

(218 words).

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[1] See Kliman (2007) and Potts and Kliman (2015) for the pitiful debate on the TSSI of Marx and Armstrong (2019a, 2019b) for a response to the deliberate mischaracterisation of MMT by mainstream economists and financial press. Armstrong (forthcoming) records how heterodox economists can agree on the mainstream's poor scholarly behaviour, with Potts, in Armstrong (forthcoming), arguing that it is time to recognise such behaviour in heterodoxy.

Introduction

In the neoliberal period (from around 1980 to the present day) the primacy of markets has been supported, indeed preached, by mainstream economics, and defined by a deep-seated mistrust of the economic competence of the state and the lauding of market forces, including privatisation, outsourcing, and ever-lengthening global supply chains. In this paper we consider how economic theory can help us understand the Covid-19 crisis and – if important lessons are learned - what this might lead to with respect to policy; we argue that *it is time for informed change*.[2]

We argue that the purpose of economic theory should be to describe the economy, warts and all, free of any agenda designed to justify what is happening (or merely reflective of what the economist would like to happen). Both idealism - specifically, as manifest in an unquestioning faith in the efficacy of markets - and ideologically-based justification of 'free-markets' detract from our ability to accurately explain what is actually happening in the economy. We, of course, accept that a full separation of economic theory and ideology may well be impossible[3], nevertheless, we argue that economists should aim to describe the nature of economic reality[4]. *Theory should start with ontology*, rather than be founded upon so-called axioms which are claimed to be objective but are best conceived as manifestations of an *ethical preference for individualism* (Smithin 2009). We contend that, without this overriding focus on 'realism', or simply, 'realism all the way' (Smithin quoted in Armstrong forthcoming) theories will be ill-designed to explain the world to which they are applied, policies will be 'tripped up' by reality or, policies might only achieve – albeit imperfectly – a secondary 'hidden agenda'– that of favouring the elites.

In this distorted sense mainstream economics' 'leave it to the market' mantra does deliver the desired outcome of favouring the 'right people'. We have no doubt that it played its part well in reducing the power of the left and the trade unions from the 1970s. However, we profoundly disagree with the suggestion it provides an 'effective handbook' to consult when seeking insights to inform policy. The stakes for society (now, with the Covid-19 crisis and - arriving like an express

[2] We believe that any post-crisis return to a misplaced faith in the general efficacy and the inherent stability of markets would merely be the result of unjustified inertia on the part of the economics profession undergirded by ideology not evidence (Potts and Armstrong in Armstrong forthcoming).

[3] Joan Robinson (1962: 140, emphasis in the original) notes, '…taking a particular economic system as given, we can describe the technical features of its operation in an objective way. But it is not possible to describe a *system* without moral judgements creeping in.'

[4] See Lawson 1997, 2003, interview in Armstrong forthcoming; Smithin 2009, Armstrong 2018)

train -the climate emergency) are too high to use theory which we argue draws upon this distorted ideology. This is why we look to MMT and Marx as theoretical bases to discuss policy; we believe we are at least strong enough to look beyond our own ideologies and to explain how these theories can best describe reality and thus inform policy.

1. Why The Fiscal Constraint Is An Illusion

In orthodox economics, the government is treated as an entity which faces a budget constraint (in the manner of a household); its spending is limited by its ability to tax or borrow (Mitchell 2011; Armstrong 2020). Mainstream thinkers do acknowledge that a third method of finance exists; that of 'money printing'. However, this means is effectively ruled out as it is argued that it *necessarily* results in inflation. The advocates of this perspective contend that money printing and bond issuance are *alternative* ways of financing a deficit. They argue that such inevitable inflation can be avoided if, instead of printing money, the state 'finances' its spending by debt sales to the private sector. Unfortunately, according to mainstream thinking, this method also involves its own problems- in particular, 'crowding out'[5]. Thus the government's spending capacity is constrained; it is essentially 'hemmed-in' and its ability to utilise fiscal policy is thus limited.

In contrast, the advocates of MMT conceptualise the state as a currency-issuer. From this viewpoint, in reality, bond issuance and money printing are not alternatives. The insights of MMT support the argument that the government *always* spends by the creation of new money and that both taxes and borrowing logically and historically follow spending (or lending). For Modern Monetary Theorists, only money that has already been issued by the state can be collected in taxes or used to buy state debt. When the government spends it does so by crediting the bank accounts of its target recipients, simultaneously increasing the target's bank's reserve account by the same amount. When taxes are paid by a private sector agent her deposit balance falls and her bank's reserve account balance at the central bank (CB) is correspondingly marked down[6]. The purchase of government debt (by private agents or the CB) is best conceptualised as a *reserve drain* (Mosler 2012; Armstrong 2020, 2015) which changes the composition of non-government sector holding of risk-free state debt but not its size.

In reality, the government sets its aims, determines its budget and spends by the *ex nihilo* creation of new money. So, if significant unemployment exists there is no barrier to increased government net spending to reduce it, there is no fiscal constraint, irrespective of the size of the government

[5] The crowding hypothesis is based on the contention that higher interest rates will lead to lower private sector investment, meaning that large government deficits effectively 'crowd out' private investment. Little, if any, empirical support for this hypothesis exists (Armstrong 2015).

[6] It is important to stress that private sector debt or bank money cannot provide the *final* means of settling a tax bill which occurs when a taxpayer's bank's reserve account at the central bank is debited in favour of the Treasury account (Armstrong 2019a, 2019b).

deficit both in absolute terms and as a proportion of national income. Only in an economy suffering from inflation would fiscal contraction be a potentially appropriate option (Armstrong 2020). We are not suggesting that MMT's conceptualisation of a government as a currency–issuer provides the answers to all the questions of macroeconomics – far from it – we merely argue that it provides a superior framework within which to analyse economic questions (which are social questions with more than one solution). The advocates of MMT never argue that the government of a nation with its own currency under floating exchange rates faces no constraints, they argue that these constraints are real, not monetary. The nature of real resource limits creates the possibility of inflation, whether that be through 'overheating' following from scarcity, imperfectly categorised as demand-pull inflation, or through downward exchange rate movements or as the result of social conflict/distributional issues, imperfectly categorised as cost-push inflation.[7] The problem of mainstream macroeconomics is that it is stuck in the world it was built for, one where unemployment is regarded as an equilibrium market outcome to control of inflation. Mainstream economics provides no framework within which to develop an understanding of the multiple possible causes of inflation and the related idea that different inflationary pressures could be dealt with in different ways, each with different social consequences. Expansionary fiscal policy may or may not lead to significantly inflationary pressure – depending on the cause of the inflation.

Assessing the real cost of state purchases involves a consideration of the alternative uses to which the resources drawn into a particular employment by government spending could have been employed. If the states decides to employ 20000 more nurses then the real cost to the nation is not the money required to do so but rather it is the loss of their potential output-generating work in the sectors within which they might otherwise have been employed. We suggest that Modern Monetary Theory provides a superior framework within which debates about such issues as the size of NHS, the education service and the army can be conducted. Such questions are, in essence, political but we argue that the most effective way to seek for answers necessarily entails the adoption of an underlying framework which recognises the real costs and benefits and choices rather than a mistaken focus on state debt and deficits – which - in of themselves - are of no importance.

[7] We should not forget the role of taxes and interest rates directly play in effecting prices and wages.

2. Central Banking and Monetary Policy - A Battle Between Ideology and Reality

Alongside their conceptualisation of the government as a currency-user, orthodox economists – implicitly or explicitly - accept that money is neutral (at least in the long run). We argue that the relationship between theory based upon this foundation and the monetary policy enacted by central banks has become increasingly tenuous since the 'heyday of monetarism' in the late 70s. In other words, a 'disconnect' exists between mainstream theory and the conduct of monetary policy; a 'disconnect' between theory as ideology and what really happens in terms of policy. We contend that, in general, this 'disconnect' has been widening, as periodically manifested in a series of central bank rejections of the tenets of monetarism, such as the exogeneity of the money supply and the validity of the quantity theory of money. However, central bankers still reassert the validity of a belief in the long-term neutrality of money, or at least word their heresy carefully so as not to directly contradict macroeconomic orthodoxy in public pronouncements. Intriguingly, Goodhart (personal correspondence 2020) notes that he does not consider that the underlying attitude of central bankers towards money exogeneity and money neutrality has actually changed much in recent decades,

> Unlike some economists, they never thought that money was exogenous. They always knew that what they were doing was setting the official short-term interest rate. Given the interest rate, the determination of money stocks, both broad and narrow, depended on such factors as the public's demand for credit, etc. Again, they thought that over the foreseeable horizon monetary policy would have real effects. The longer into the future, the more the outcome would fall on the price level and the less on output, but the idea that money would be fully neutral at some distant horizon was always more a matter of theory than a practical issue for central bankers.

It could be argued that the failure of the so-called 'Volcker experiment' might be considered as an early major retreat from monetarism. In 1979, consistent with monetarism, when Paul Volcker was chairman of the Fed, the intermediate target of choice was changed to the rate of growth of M1 (Wray 1998). Implicit in this approach was the need to permit the federal funds rate to rise to a level consistent with achieving this aim. However, as noted by Wray (1998: 101), such a policy was doomed to fail from the outset.

> The Fed would calculate the total reserves consistent with its monetary target then subtract existing borrowed reserves to obtain a non-borrowed reserve operating target. If the Fed did not provide sufficient reserves in open market operations (as it hit its non-borrowed reserve target), banks would simply turn to the discount window, causing borrowed reserves to rise (and, in turn, the Fed to miss its total reserve target). Because required reserves are always calculated with a lag the Fed could not fail to provide the needed reserves at the discount window. The Fed found it could not control reserves.

Conventional thinking is based on the idea that banks need excess reserves before making loans and a 'deposit multiplier' exists. However, in practice, banks make loans without reference to prior reserve positions (Mosler 2012: Armstrong 2015, 2020). Faced with a customer deemed credit-worthy, a bank makes the loan and obtains the reserves later. Thus, the quantity of reserves is determined by the amount of loans issued, not the other way around. In this case, the role of the Fed is passive. At this point it should have been all too apparent that monetarism, in its 'pure' form, is simply a utopian project which cannot work in practice; it would require wild swings in interest rates at best and the breakdown of the interbank payments system at worst. Clearly, such a situation would be politically unacceptable; this was reflected in the conduct of the so-called Medium Term Financial Strategy (MTFS) in the UK where Thatcher's government emphasised the free market dogma and inflation-prioritising of monetarism while keeping interest rates under their control for political purposes, rather than attempting to control the level of reserves in the banking system (Armstrong 2015).

The failure of monetary targeting left central banks facing a dilemma; accepting short term money neutrality and using 'pure' monetarism would be a practical disaster but the rejection of mainstream theory was also off the menu. In this quandary, New Keynesianism proved very useful (and continues to do so) as it accepts the short-term non-neutrality of money, legitimising a central bank's use of the short-term interest rate to influence not only inflation but also real variables. Crucially though, New Keynesianism retains the view that money is neutral in the long run and therefore merely determines the price level (Armstrong 2015). From the perspective of central banks, the attraction of this 'compromise nature' of New Keynesianism is strong; as was reflected in this Bank of England statement from 1999.

> Rather, in the long run, monetary policy determines the nominal or money values of goods and services … However, monetary policy changes do have an effect on real activity in the short to medium term. And though monetary policy is the dominant determinant of the price level in the long run, there are many other potential influences on price-level movements at shorter horizons. (The Bank of England 1999).[8]

After the Global Financial Crisis (GFC), central banks enacted unconventional monetary policy tactics, in particular Quantitative Easing (QE, called Asset Purchases in the UK when carried out by the Bank of England). The application of this policy initiative resulted in increases in the money supply (as conventionally defined)[9]. In order to reassure financial institutions and the public at large, the Bank of England was effectively forced to admit (and repeatedly reiterate) that the resulting significant increases in the quantity of money would not be inflationary, in other words, they had to cast doubt upon the validity of the quantity theory of money in some cases and in others even actively repudiate it. We might consider two speeches from 2009 by way of illustration, First, David Miles[10] made this point in a speech on 30 September 2009.

> *However, it is important to realise that there is very little economic content in MV=PT, and there is no message from it about economic behaviour. It is a bad mistake to think that this relation is analogous to, say, another more famous equation: E=MC2, which almost certainly embodies some profound truth about the universe.* (Miles 2009, parentheses in the original, emphasis added).

Second, Adam Posen[11] noted the following in his in speech on 26 October 2009. 'I hope to shift the focus away from some concerns that are unfounded, like the idle chatter one sometimes hears about printing money causing inflationary periods' (Posen 2009).

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[8] This 'compromise' view of the neutrality or non-neutrality of money is still advocated by the Bank of England in 2020, certainly in their 'official' discourse (see Bailey 2020 below).

[9] Since government bonds are not included in the money supply, debt purchases will add to the money supply by adding reserves.

[10] David Miles is Professor of Financial Economics at Imperial College Business School. Between May 2009 and September 2015, he was a member of the Monetary Policy Committee at the Bank of England.

[11] Adam Posen is President of the Peterson Institute for International Economics. From September 1, 2009 to August 31, 2012, he was a voting External Member of the Monetary Policy Committee of the Bank of England, by appointment of the UK Chancellor of the Exchequer.

A further rejection of orthodox economic theory occurred in 2014, when the Bank of England criticised the orthodox view of banking; rejecting the ILF (intermediation of loanable funds) model of banking which conceptualised banks as pure intermediaries between savers and borrowers. In two key articles (McCleay et al. 2104a, 2104b), the Bank explained that banks can *create* money (in the form of bank deposits) and do so whenever they make loans. This admission lent support to the endogenous approach to money – as opposed to orthodox monetarist arguments in favour of the exogeneity of money - which had been advocated by heterodox economists (especially Post-Keynesians) for many years (Wray 1990, Armstrong 2015).

The recent Covid-19 crisis has seen further evidence of a move away from orthodoxy on the part of the Bank - its nature is somewhat different since it involves fiscal policy on this occasion. The huge demand shock associated with the Covid crisis, greatly exacerbated by the ensuing lockdown, effectively forced the government to spend 'whatever it takes' to prevent economic collapse[12]. In 2020, the Bank admitted (Bank of England, April 9, 2020) that the government *can spend whatever it decides to,* irrespective of the size of its balance at the Bank. However, the admission was tempered by an attempt to deny the government's ability to spend without 'permission' from its own Bank and assert that the 'overdraft' would be repaid 'as soon as possible'.

> HM Treasury and the Bank of England (the Bank) *have agreed to extend temporarily* the use of the government's long-established Ways and Means (W&M) facility. As a temporary measure, this will provide a short-term source of additional liquidity to the government if needed to smooth its cash flows and support the orderly functioning of markets, through the period of disruption from Covid-19.
> The government will continue to use the markets as its *primary source of financing, and its response to Covid-19 will be fully funded by additional borrowing through normal debt management operations. Any use of the W&M facility will be temporary and short-term*. As well as temporarily smoothing government cash flows, the W&M facility supports market function by minimising the immediate impact of raising additional funding in gilt and sterling money markets. The W&M facility is the government's pre-existing overdraft at the Bank. Any drawings will *be repaid as soon as possible before the end of the year*. HM Treasury, the Debt

[12] 'We are in an entirely new world. A wartime effort, with wartime deficits to cover it', Rishi Sunak, quoted in Islam, F., BBC News online, 17 March 2020.

> Management Office and the Bank will continue to cooperate closely to support the orderly functioning of the gilt and sterling money markets.
> (Bank of England April 9, 2020, parentheses in the original, emphasis added).

Looking more deeply, this is what we interpret the Bank to be saying; first, its statement shows that it accepts (at least in public pronouncements) that the government faces a budget constraint (GBC) and in its pronouncements the Bank attempts to make the government's current greatly increased spending 'cohere' against this backdrop belief. The Bank has framed its statement in a way that suggests that, although it accepts that debt sales are not presently required, as soon as is practical, 'normality' will return and government bonds will be sold to 'finance the spending' which is now - temporarily at least - accounted for as an overdraft. Second, The Bank's approach shows that - despite the denials of the tenets of monetarism described above – it still tacitly supports the idea of the long-term neutrality of money (or appears to do so in the recent public announcement).

The latter view is apparent from the Bank's professed unwillingness to allow the additional reserves created by the state spending to remain in the banking system as they constitute an increase in the stock of money (conventionally defined) and will have (at best) long-term inflationary consequences. The Governor of the Bank made the point that 'Using monetary financing would damage credibility on controlling inflation by eroding operational independence. It would also ultimately result in an unsustainable central bank balance sheet and is incompatible with the pursuit of an inflation target by an independent central bank' (Bailey 2020).

However, from the perspective of MMT such equivocation is profoundly mistaken. In this crisis nothing has changed; the government *always spends by the issue of new money* and all that has happened is that – temporarily at least – the government has been released from its unnecessary, and indeed, voluntarily-accepted requirement to sell debt to match its deficit. In other words, the government is apparently no longer required to 'recapture' previously issued state spending as credits in its account at the Bank by debt sales before spending. From an MMT perspective, such a debt sales policy should never be described as 'financing' but merely as a voluntary asset swap. The government is the monopoly issuer of the currency and is therefore never *operationally required* to sell debt to match its deficit. It can leave the reserves in the banking system. The result of such a policy will be that the overnight rate will fall to zero (unless action is taken, such as paying interest on excess reserve balances held at the central bank). There will be no effect on

inflationary pressure in the short or long term (Mosler 2012: Armstrong 2015, 2020). If the government fails to sell debt, the amount of net saving of the non-government sector remains the same – just its composition in terms of risk-free assets (reserves or gilts) – is different from the case when debt is sold.

In summary, we might note that the Bank's inability to fully commit itself to the non-neutrality of money in both the short and long run, combined with its apparent inability or unwillingness to admit that the government's status as a monopoly-issuer of the currency (and the complementary insight that it cannot be operationally required to sell debt) mean that the Bank is effectively forced to rehearse the line that the government – sooner or later – will need to 'fund' a deficit by debt sales. The Bank as yet does not understand, or at least acknowledge as Mosler (2011) notes, 'It must be impossible for the Fed to create inflation.'

The point MMT has concretely shown us is that if the government engages in deficit spending the impact on the economy will be the same – irrespective of whether bonds are sold, either to the central bank or non-government sector, or indeed no bonds are issued, and the deficit is recorded as an 'overdraft' at the Bank (essentially, the latter should be considered as merely an alternative means of internal accounting for the monetary flows between the state and its Bank).

3. Banking Concretely for People and Business.

MMT helps us understand that a policy, such as QE, which injects excess bank reserves into the banking system will drive the overnight risk-free interest rate to zero, unless action is taken, usually the central bank agreeing to pay a 'floor rate' on excess reserves (Mosler and Armstrong 2019). Conventional analysis suggests that, in normal times, interest rates may be manipulated to speed up, or slow down, the economy, however, it fails to grasp the *actual effects of interest rate changes*. Central bankers believe raising rates is contractionary and reducing rates is expansionary (see, for example, Saunders 2020). However, Mosler and Armstrong (2019) and Armstrong (2019b) consider the effect of the state as a *net-payer of interest* to refute this idea.[13]

Interest can be a significant cost to firms and when interest rates are set sufficiently high in a crisis, monetary policy can be disastrously 'effective' in its effects upon the profitability (or even viability) of firms and the financial system itself (Marx, 1981, Chapter 34). However, low rates will not stimulate expansion. What is also clear is that, as Marx terms it, fictitious capital (any security representing the capitalised value of an income stream) will depend on the short-run risk-free interest rate. Marx (1981: 595, then 597-598)

> The form of interest-bearing capital makes any definite and regular monetary revenue appear as the interest on a capital, whether it actually derives from a capital or not…
> The formation of fictitious capital is known as capitalization. … For example, if the annual income in question is £100 and the rate of interest 5 per cent, then £100 is the annual interest on £2,000, and this £2,000 is then taken as the capital value of the legal ownership title to this annual £100… Moving from the capital of the national debt, where a negative quantity appears as capital – interest-bearing

[13] A change in interest rates redistributes income between savers and borrowers. CB orthodoxy suggests that, given that propensities to consume are taken to be higher for borrowers than savers then, for example, a rate rise will be contractionary as savers will increase their spending by less than borrowers reduce theirs. 'However, although the propensity estimates of the central bankers may well be accurate, given the state is a net payer of interest to the economy, higher rates are adding interest income to the economy and lower rates are removing interest income from the economy. With debt to GDP ratios often approximating 100% of GDP, the interest added or subtracted by this channel is likely to dwarf the effect of the differing propensities between private sector borrowers and savers. Lower rates may help borrowers to service loans and qualify for new loans, but lower net income works against new borrowers' income levels and the general ability to service loans in the economy. Thus, higher rates are in fact an expansionary force rather than the contractionary force assumed by central bankers' (Armstrong 2019b).

> capital always being the mother of every insane form, so that debts, for example, can appear as commodities in the mind of the banker.

In crisis, as income streams are threatened, fictitious capital can lose value, but very low interest rates limit this loss. Very low interest rates in crisis are thus concretely supportive of holders of financial assets, but only a mythical green light to unstoppable inflationary boom in the real economy. 'Real' borrowers in crisis, whether the general public or small (or 'micro') businesses, do not think, 'let's borrow more as the interest rate is low' (unless they have incredibly secure income)! More formally, the interest elasticity of demand for loans is very low so borrowers just seek to pay their risk-adjusted interest rates and meet due repayments of principal. If they borrow more, even at escalating risk-adjusted rates, it is in panic to avoid default (Marx 1981: 735, then 745).

> Usury proper not only continues to exist, but in countries of developed capitalist production it is freed from the barriers that former legislation had always placed to it. Interest-bearing capital retains the form of usurer's capital vis-à-vis persons and classes, or in conditions where borrowing in the sense appropriate to the capitalist mode of production does not and cannot occur; where borrowing results from individual need, as at the pawnshop; where borrowing is for extravagant consumption; or where the producer is a non-capitalist producer, a small peasant, artisan, etc. … finally where the capitalist producer himself operates on so small a scale that his situation approaches that of those producers who work for themselves … the renting of houses, etc. for individual consumption. It is plain enough that the working class is swindled in this form too, and to an enormous extent … This is a secondary exploitation, which proceeds alongside the original exploitation that takes place directly within the production process itself.

By Marx's definition, we 'little' people are subject to usury through our overdrafts, credit cards and mortgages (Potts 2011). Unlike firms that can make a profit (by extracting surplus-value from their workers)[14] and share their profit with the 'monied' capitalist, we little people have no (or very few) workers to exploit, so are in danger of losing our possessions to the usurer. But this is a 'side-show' to capitalism proper, i.e. capitalist production will not be held back if this 'primitive' business

[14] As we shall consider below this process to Marx is in no means as direct as it sounds here, even though this is the essence of the matter.

is reformed; indeed, the limits to financialisation are brought to the fore by Marx's value theory. To Marx, *finance is unproductive* (Marx, 1978, Chapter 6); it is a drain on surplus-value extracted in production, while further secondary exploitation is a usurious drain on little people.[15]

Our concrete problem is that the finance required by little people is overwhelmingly in the hands of our dominant banks. This dominance has been cemented by the rundown of social housing and explosion of mortgage lending; little people are big business to finance in the UK. Our analysis drawn from Marx argues for a form of capitalism with social housing as the norm, contending that it would be free of the main problem of usury and would be more effective in *capitalist terms* for this.

We can conclude that, when it comes to policies to move to state banking for little people, the agenda is clear, the objective must not be to become an alternative usurer, but to support socially-desired outcomes. If bad loans are incurred they can be written off by the state, normal calculation of risk-adjusted interest rates need not apply. For a local government, bank lending to local business should be a social decision. The key point of financial 'prudence' is the decision to grant the credit in the first place. But any significant reform of 'little people finance' must address the question of housing. Lack of a plan (or alternatively, the call of free-market small-minded greed) has led us into our unaffordable housing stalemate. A fall in house prices would hurt, but be good for both the economy and, ultimately, workers' ability to reproduce themselves.

Society has a choice if it wants to try and help little people, it might prop-up (or subsidise) the current usurers, and let them continue to hold the economy back in the long-term, or embrace radical reform and let house prices fall, and to the extent people become bankrupt directly, provide help (by buying out mortgages and converting to social ownership), and re-invigorate the housing market by purchasing existing houses and building new houses for social ownership. This is radical but, by Marx's understanding of capitalism, is good for capitalism itself.

But what of business beyond the small (or micro level); what is it best here for society to do? The question is far from straightforward, but again we suggest Marx helps us posit an answer. For Marx, in any industry there are leading, average and laggard producers, and any producer is

[15] If workers' secondary exploitation is ultimately, at least in part, paid for by higher wages to cover the higher reproduction cost usury imposes on workers, then secondary exploitation holds back primary exploitation in production.

highly unlikely to actually appropriate through the market (the formation of price) the value they extract from their own workers in production.[16] Leading producers are leading producers precisely because of their ability to appropriate above-average profitability. So, when a crisis, such as the Covid crisis, threatens to temporally disrupt leading producers' profits, even cause losses, their likely future high profitability will allow them to avoid problems of rolling over loans at escalating risk-adjusted interest rates. They will be able to borrow more or to attract capital by conducting rights issues. To defend the interest of their current shareholders (who don't want future dividends held back by higher borrowing costs or diluted by having to share profits with new shareholders), these leading producers may apply for cheap loans, or even grants, from the government[17]. If they succeed, society is simply subsidising the most effective capitalists, and, even more precisely, a particular group of existing shareholders of those leading capitals.

Private finance can, at least in theory, save the leaders, but what about the rest of business, can private finance help, and what should the state do? Business failure often leads to financial restructuring (through bankruptcy or takeover) with the underlying business continuing. Unemployed resources can be employed by new entrepreneurs. The collapse of individual capitalists is a typical part of capitalism, where change is normal. For example, the high street is in danger and commercial property prices are inflated. If the Covid crisis bankrupts many retailers and depresses commercial property prices, then entrepreneurs of many forms, both private (seeking profit) and public (pursuing social objectives), can take advantage of cheaper costs to innovatively re-invigorate city centres.

[16] To Marx, for the economy as a whole, total profit is constrained to total surplus-value extracted from productive workers, minus consumption of surplus-value by unproductive activities, such as finance. Monopolies, shielded from competition, take their share of total profit, leaving the rest to be competed for in the competitive economy. As Marx (1981, Chapter 9) explains, price/appropriated value tends to deviate from value/produced value across industries to equalise profitability per unit of capital applied (transferring surplus-value to industries that employ much constant capital - machinery, raw materials - as compared to living workers - the source of surplus-value), from labour-intensive industries. Furthermore, within each industry we have leaders (tending to be larger with lower unit costs), and laggards. Assuming an industry had, in aggregate, average profitability (and the economy had no monopolies or unproductive consumption of surplus-value), so it appropriated in profit the surplus-value it appropriated from its workers, then only a completely average firm in that industry would appropriate the same surplus-value as it extracted from its workers. Leaders would appropriate more value than they produce, having higher profitability, at the direct expense of laggards appropriating less value than they produce, having below average profitability.

[17] Bank of England support is indeed available to selected firms; https://www.bankofengland.co.uk/markets/Covid-corporate-financing-facility

Trying to keep everything the same is not capitalism - or appropriate, socially - given our environmental challenge. Do we really want to save all our airlines and international travel businesses for now, and then think about the environment *later*?[18] In capitalism, we adjust by crisis, not central planning, we do not have the social system to guarantee private employment but we do have the fiscal space to pay unemployment benefits, *and* to introduce job guarantee (JG) schemes (see below), *and* to support the public sector, *and* to expand social housing. We can spend on 'greening' our energy system, transport, conversion to electric cars, cycle paths; we can do all this and more, but preserving an environmentally unstable status quo is not good for society or capitalism. Neither is standing in the way of bankruptcy and falling fictitious capital values good for capitalism, as this is the very way Marx thinks that capitalism reinvigorates itself by resetting the conditions for a higher profit rate, as we shall explore below.[19]

In summary, it would be better for society to increase unemployment benefits than to financially support weak or socially undesirable capitalists. For these reasons, we see little role for the state to prop up capitalists, indiscriminately. The public would be better served if firms go into bankruptcy and are financially restructured, even if it is to be the public sector that buys them in the end. Our banking system may be so dysfunctional, obsessed in its usurious ways, that there is a role for the state as a banker, charging 'reasonable' risk-adjusted interest rates[20]. In this sense, the creation of state banks would be *efficient*, but such an aim is too limited. State banking should be democratic and local. It makes sense for the local government-controlled bank to lend to local, often small, business at subsidised (by the state paying for write-offs) rates focusing on environmental and social objectives. Again, this is likely to be a far better use of public money to support people rather than lending to prop up large capitalists that can be easily reinvigorated by financial structuring.

While fictitious capital would be falling in price, the state would be acquiring assets in social housing, loans to business and businesses themselves. People will become unemployed, but there is no need for lasting unemployment, given the state's position as a monopoly-issuer of the

[18] Sealy (2018) explains just how zero dollar for the local economy industrial scale international tourism is in the Caribbean, and at the same time an environmental disaster, with even the fruit being imported from far away.

[19] Private/employer pensions may be affected, but pensions are very unequal in the UK, with the state always having the fiscal space to pay decent state pensions, whether they tax rich pensioners or not.

[20] This is not to say private banks could not operate. A private bank could conduct business, provided it operated efficiently within the regulative guidelines. Effectively, the state bank puts a competitive 'floor' on banking quality. Private banks would be required to match, or better, this standard to prosper. Such an approach encourages innovation -provided such innovation serves public purpose as well as private interests.

currency, it can always set the level of unemployment at zero by utilising a JG scheme (Wray 1998; Mitchell 2011; Mosler and Silipo 2016)[21]. Low risk-free interest rates are likely to accompany excess bank reserves for some time, but this is no consequence to capitalism as a whole; the monied capitalist is simply taking less to the advantage of the functioning capitalist, (Marx (1981: 482-484), 'the circumstances that determine the magnitude of the profit to be divided, the value product of unpaid labour, are very different from those that determine its distribution among these two kinds of capitalists, … The prevailing average rate of interest in a country, as distinct from the constantly fluctuating market rate, cannot be determined by any law. There is no natural rate of interest.'

It is a social matter in capitalism, among the family of capitalists, how to split the profit, that modern central bank practice has turned into a public decision, which we have been 'extremely' making ever since the 2008 crisis. In other words, for Marx, there is no *normal* to return to.

[21]If the state introduces a JG scheme it offers a job, at a fixed wage, to all those ready and willing to work, effectively reducing unemployment to zero. *Such offers are entirely voluntary so a JG should never be confused with 'Workfare' or other similar schemes which are either compulsory or required as part of a contractual obligation for those in receipt of benefits.* Those unwilling or unable to work are not classified as 'unemployed' and would not fall within the remit of a JG scheme (Wray 1998).

5 Can We Efficiently Exploit Ourselves from Home?

Marx (1976, Chapter 7) defines capitalism as a dual process, the labour process and the valorisation process. The labour process is how 'real' resources are put together to produce real outputs (use-values); all societies must have labour processes. The second part of capitalism, its sole purpose, unique to capitalism, is making a profit through expanding the exchange-value of capital. Although individual capitalists do not actually appropriate the surplus-value they exploit from their own workers, for any level of surplus-value they do appropriate, the more they can reduce their variable capital (wage) cost per commodity the higher their profit will be. It pays to extend the working day and also to increase the intensity of labour.[22]

Clearly, some labour processes cannot be conducted at home in any society, like building a bridge. But in capitalism many labour processes, that could now be done remotely given the technology available, are still done in the workplace to allow (in-of-itself unproductive) supervision to enforce the intensity of labour.[23] So people may be called back to work prematurely when they can work at home purely to ensure they work sufficiently hard. We have long seen how the building industry's traditional supervision has often been effectively replaced by contracting out, in essence, re-establishing piece work. The spread of this form to the creative industries is concretely reported by Pitts (2019). The 'gig' economy is propelled, indeed defined by, self-exploitation. So, we should beware the likely long-term consequences of working from home if we expect to be salaried, 'enjoying' employment rights and effectively represented by a union. In the short run, to the extent we might work less intensively at home then we are either 'volunteering' to extend our hours to get the same amount of work done for the same pay, or being less effective, and thus more expendable to our employers. Taking a rest in capitalism's time, and not our own time, is not an option in capitalism.[24] Our rights to free time, just as our wages (our cost of social reproduction), are determined by class struggle, and given that struggle is a collective endeavour, it is made harder by being isolated at home.

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[22] For Marx (1976, page 987) an hour of abstract social labour is an hour of 'unskilled'/simple labour preformed at average intensity of effort (with skilled labour added more value in portion to its higher reproduction cost, Marx, 1976, page 135). If all work 'harder' average intensity rises, more commodities are produced per hour of labour, just as productivity growth in general produces more commodities per unit of abstract social labour. To produce more value the capitalist must increase the intensity of their workers' work in relation to the average.

[23] For Marx (1976, page 988) expenditures of labour necessary to the supervision/co-ordination of the labour process, no matter in which society it may be performed, are productive. But additional supervision aimed at simply maximising the self-expansion of capital in the valorization process i.e. working all as hard as possible, is not productive.

[24] One of us was granted by our employer 'well-being' Fridays off in May, but as there is no adjustment to the tasks expected to be completed, or when they are to be completed by, the gesture is utterly unhelpful and cost-less to our employer.

6 Looking Further Afield

(i) How Open Should or Can We Be?

Keynes explained in 1933 how economists have a deep-seated cultural faith in 'free trade', 'I was brought up, like most Englishmen, to respect free trade not only as an economic doctrine which a rational and instructed person could not doubt, but almost as a part of the moral law. I regarded ordinary departures from it as being at the same time an imbecility and an outrage' (Keynes 1933: 755). Although we do not oppose international trade *per se* or advocate protectionism, as conventionally defined, we nevertheless argue that increasing domestic production, where feasible and efficient in terms of real resource use, is surely beneficial. Again, this point was developed by Keynes in 1933,

> A considerable degree of international specialization is necessary in a rational world in all cases where it is dictated by wide differences of climate, natural resources, native aptitudes, level of culture and density of population. But over an increasingly wide range of industrial products, and perhaps of agricultural products also, I have become doubtful whether the economic loss of national self-sufficiency is great enough to outweigh the other advantages of gradually bringing the product and the consumer within the ambit of the same national, economic, and financial organization. … a moderate increase in the real cost of primary and manufactured products consequent on greater national self-sufficiency may cease to be of serious consequence when weighed in the balance against advantages of a different kind. National self-sufficiency, in short, though it costs something, may be becoming a luxury which we can afford, if we happen to want it (Keynes 1933: 760).

> The decadent international but individualistic capitalism, in the hands of which we found ourselves after the war [WW1], is not a success. It is not intelligent, it is not beautiful, it is not just, it is not virtuous - and it doesn't deliver the goods. In short, we dislike it, and we are beginning to despise it. But when we wonder what to put in its place, we are extremely perplexed (Keynes 1933: 760-61).

> I sympathize, therefore, with those who would minimize, rather than with those who would maximize, economic entanglement among nations. Ideas, knowledge, science, hospitality, travel - these are the things which should of their nature be international. But let finance be primarily national (Keynes 1933: 760).

Although we, like Keynes, see a case for being more local we think that it is important to recognise how important international markets are to leading producers. Marx is clear, as leaders advance, they tend to capture larger and larger, often global, markets. Patenting knowledge to support research and development is central to this process.[25] For the UK, finance is a problem. It is a leading, although unproductive, producer that captures much value from abroad. But is it moral to reform finance in the UK to limit usury at home but to let finance continue to prey abroad?[26] Importing from developing countries is key to their development, but is environmentally - and again, morally - questionable if it is purely an exercise in super-exploitation of the most disadvantaged segment of the now international labour market (Marx: 1976: Chapter 25) through, effectively, contracting out simple production (contacting 'in' for global companies).

The puzzle of accessing the benefits of both 'free trade' and protectionism has led to the phenomenon of regional trading blocs such as the European Union. Selective protection is combined with large home markets for leading producers and the power in trade negotiations to provide those producers with global access. Alone, the UK may be less able to deliver such access to its leading producers, but it does have the macroeconomic policy space to reform our economy and drive growth. What is clear is that it is illogical to isolate ourselves and create this macroeconomic space only to decide not to use it owing to some anachronistic faith in 'free-trade' and free-market economics, both ideas whose time in the ascendency has surely passed if we are to be serious about the environment.

[25]Potts (2007) analyses how patents are the manifestation of the essentially unproductive (of exchange-value not use-value) nature of research and development.

[26]The Swiss do show how leading in finance can be combined with protecting domestic production, at the cost of increasing the cost of living.

(ii) What About the Majority of the World's Population?

Here we hit the reality of how unequal the capitalist world really is. According to IMF (2019) 'real' GDP per capita in 2012 $'s was $24,297 in advanced countries in 1980, rising to $43,267 in 2019, but $3,490 in 1980 in the rest of the world (excluding India and China), rising to only $4,284 in 2019 (for India rising from $654 to $1,957, and for the actually developing China rising from $732 to $9,038). By using these imperfect statistics, we can see how the advanced, through their leading producers, have largely reaped the benefits of the technological progress capitalism has stimulated over the last 40 years, as established by global prices. Freeman (1996) explains how, for Marx, it is price formation that determines exchange value and the transfer of value both within and between countries. Trade exchanges equivalent exchange values, but price formation has already caused these exchange values to deviate from their produced values, explaining the massive transfer of value or human effort which is responsible for extreme global inequality.

An alternative definition of a developing country is a country with no leading producers, left to suffer the buffeting of variable raw material prices; a precarious situation leaving developing countries prone to usury. They are unable to run current account deficits when developed nations are unwilling to net save in their currency (Mosler 2010: 59-62), meaning that if they need to import food and vital materials they must export real goods and services to 'pay' for the imports (even when the real terms of trade disadvantage them significantly). If the developing nation is real resource poor, making this option impossible, it may be forced to borrow 'hard currency' (usually, $US) from commercial banks in developed nations or from international institutions, in particular, the IMF. The usual IMF (and orthodox) preference for free capital mobility has often been at the root of developing countries opening up their economies. However, there is no evidence to support the contention that capital mobility helps economic development in developing countries (Siddiqui and Armstrong 2018). When multinational corporations see profit-making opportunities, foreign capital floods into the domestic economy, driving up the exchange rate (possibly, encouraging the state to take on extensive foreign debt and domestic elites to engage in extravagant consumption). However, when profit-making seems less likely, capital flight follows, driving down the exchange rate, often precipitously. Governments are then forced to borrow in $US to fund imports and the burden of usurious debt becomes so heavy it depresses the living standards of ordinary people who are already poor.

Support for globalisation has characterised the neoliberal period. Such an approval has been associated with the championing of free capital movement and so-called ‘barge economics’ with the necessity of global outsourcing by multinational firms in order to reduce unit costs in monetary terms (even when in ‘real’ terms production is less efficient, plus the un-costed damage to the environment) accepted as part of the system (Palley interview in Armstrong forthcoming). Such advocacy may seem sustainable when viewed through a ‘capitalist lens’, however, on ethical and environmental grounds it must surely be rejected owing to the extremely pressing need for global action against climate change. In addition, capitalism breeds poverty (Marx, 1976, Chapter 25), and the ‘have-not’s’ cannot afford the high prices (protected by patents) that are in the interest of advanced countries’ leading producers. Capitalism has reached an environmental limit; the grim alternative to some form of international redistribution (the return of stolen or transferred human effort) is grinding poverty and a reduced quality of life and environmental degradation for all but a small minority of elite capitalists. The imminent return of wide-scale international debt crises for developing countries is likely to bring this policy choice to us quickly; so will the IMF be the normal usurer, or will it be part of a new international system that provides genuine international social banking and direct assistance?

7 Could it All Turn Out Well in the End?

The answer to this question splits into the two sides of capitalism Marx defines, the labour process and the valorisation process. If reacting to the Covid crisis causes us to move to environmentally sustainable, and more local and community, labour processes, then we would have achieved a very positive outcome. We have mentioned policies to help society adjust in this direction, with the central point, as clearly explained by MMT, that we have the fiscal space to do it. It is hard to imagine how this change can occur without it hurting some people, but that hurt can be limited if we reform finance and our housing market, while paying decent social security and implementing a job guarantee scheme.

The unknown to this 'expansion' is not if it can 'work' but, once recovery is underway, how big will the state be? This will depend upon how strongly private sector capitalism grows. In the 'Golden Age' after WW2, private sector growth was strong and Armstrong *et al* (1991) and Kliman (2012) show private sector growth was stimulated by high profitability. So, if profitability was high post-Covid-crisis, high private sector growth would likely lead to a reduction of direct state involvement in the economy.

We have now moved to the valorisation process, making a profit. Marx (1976: Chapter 25, and 1981: Part Three) explains how normal capital-intensive growth tends to reduce the profit rate.

$$r = s / (c + v)$$

s = surplus-value extracted from productive workers
v = variable capital, workers' wages
l = s + v, total living labour input
c = constant capital, all other inputs, raw materials, machines etc.

As *c* tends to grow faster than *l*, and crucially *s*, in boom the rate of profit tends to fall.[27] Kliman *et al*, (2013) explain what Marx means by this tendency which he calls his most important law

[27] Kliman (2007) explains how most 'Marxists' do not think this tendency holds through using simultaneous valuation (Okishio, 1961) or employing sequential but replacement-cost valuation (Laibman, 2000). However, as Freeman and Carchedi (1996), Kliman (2007) and Potts (2009) all show, interpreting Marx as having a sequential single-system approach confirms the tendency for the profit rate to fall. TSSI of Marx scholars have argued (and have statistically

(Marx, 1981, page 319). Boom reduces the profit rate, so periodic crises are absolutely necessary to restore the profit rate. In the general case, this potentially involves an increase in the exploitation of workers, but if usurious finance and housing are reformed, both workers and productive capitalists can gain as secondary exploitation is greatly reduced. But, the restoration of the profit rate rests most significantly on the devaluation of existing capital (Marx, 1978: 250, its 'moral' depreciation by its falling in price). Many old capitalists will go bankrupt, so real capital will be cheap, and as shares, or fictitious capital, fall in price much real capital can be bought by taking-over or rescuing firms. This is precisely why we should not seek to protect *all* capital, and worst of all attempt to protect the value of fictitious capital. Fiscal space is best used for change and not for protecting individual wealth-holders at the expense of the health of the system and, indeed that of the planet.

So, the Covid crisis may be just what the capitalist system needs, but can the system recover by itself, or does it need the state to re-order production and initially stimulate growth (Freeman, 2016) This may be an interesting 'academic' question, but we cannot wait for an answer, as for the sake of both people's livelihoods and the environment we need the state to utilise its currency-issuing power to initiate recovery immediately.

shown, most extensively in Kliman, 2012) that low profitability has persisted since the end of the Golden Age, as crisis has not been strong enough to reset the profit rate. Potts (2013) argued that this Keynesian inspired stagnation was likely to persist.

8 Conclusion

We do not advocate ‘leaving it to the market’, but recognise that both ‘market’ forces and judicious state support are key to change and a strong recovery. If Marx is right, if the recovery is strong it will, in the end, reduce the profit rate and lead us back to crisis, as *capitalism is inherently unstable*. However, such an admission does not mean that the post-Covid manifestation of capitalism cannot be an environmentally sustainable capitalism, freed of unnecessary usury, with workers both better off and better protected than before. Furthermore, developed nations must play their part in building a new international capitalism free of usury, with redistribution blunting the inequality capitalism generates, as people and the environment exist everywhere. It is about building a different future using capitalism’s ability to change through crisis.

Of course, as Marx (1981: 959) hoped, we could transcend to a better system than capitalism (Potts, 2001, lays out Kalecki’s ideas on democratic rational central planning), but, for now, let us use the Covid crisis to argue for at least a ‘better’ capitalism, one that is more effectively managed by governments who are aware of the fiscal space that exists when they operate under floating exchange rates (Armstrong 2105, 2020) and, importantly, are also cognisant of the ineffectiveness of monetary policy as a tool to control inflation. Finally, let us stress again that we should eschew protecting elite interests by the indiscriminate saving of existing capitalists and wealth holders, irrespective of their contribution to the real economy and their impact upon economic efficiency and the wider environment; this is not just bad for people, it is bad for capitalism itself.

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